

Practical Problem Solving

Software Engineering

Understanding what would make things better

Gregory S. DeLozier, PhD

“We need a volunteer website”

You can already imagine how to build it.

What would you ask first?

What is actually going wrong?

A campus club staffs a table in one-hour shifts.

Nobody arrives for the second shift.

The organizer stays and misses another commitment.

What happened last time?

Follow the missed shift

What happened

A member offered to work

The member later changed the reply

Another member asked about the shift

What the organizer saw

Copied the offer into a personal list

The old commitment stayed on the list

The question went unanswered

Where did their understanding come apart?

Who has the problem?

Volunteer

“I asked, but never heard back.”

Wants a commitment they can plan around

Organizer

“I was waiting for a definite yes.”

Wants a schedule they can depend on

Ask someone to show you how they signed up.

What would make things better?

The volunteer knows which shift they agreed to work.

The organizer sees the same commitment.

A cancellation reaches someone who can respond.

Would a new calendar page achieve that?

The table was fully staffed

Good news.

The organizer also called every volunteer and reconciled the lists the night before.

What improved? What work is still hidden?

What is getting in the way?

The next event is in **ten days**.

There is **no budget** for a paid service.

Members already use the chat.

What could help in time for this event?

What are we assuming?

Belief	A way to check
Members can open the schedule	Try it on their phones
The missing volunteer forgot	Ask what happened
There are enough volunteers	Compare availability with the shifts

Which wrong assumption would waste the most work?

A tool cannot fill every gap

Eight one-hour shifts need coverage.

Members can offer only four hours in total.

What choices does the club have?

What could we try?

One shared schedule.

Requests stay in the existing chat.

The organizer records and confirms each commitment.

Try it with two members before the event.

Watch what happens

Can they find the schedule?

Can they tell when they are expected?

What do they do when their plans change?

Where did they need help?

Work you can enjoy

Understand what someone expects before committing.

Make a small improvement you can see helping.

When it fails, investigate what was misunderstood.

Clearer expectations mean less avoidable frustration.

Try a situation you know

Choose a recurring difficulty from a club, job, or ordinary part of your day.

Describe the last time it happened.

Who had difficulty? What were they trying to do?

Choose a useful next step

What would make things better?

What limits you, and what are you assuming?

What could you try? How would you tell if it helped?

Find something out before committing to the build.